PHYSICS – BOARD QUESTION PAPER : MARCH 2022

Time: 3 Hours

Max. Marks: 70

General Instructions:

The question paper is divided into four sections.

1. Section A: Q1 contains Ten multiple choice questions. Q2 contains Eight very short questions.
2. Section B: Q3 to Q14 contain Twelve questions. Attempt any Eight.
3. Section C: Q15 to Q26 contain Twelve questions. Attempt any Eight.
4. Section D: Q27 to Q31 contain Five questions. Attempt any Three.
5. Use of log table is allowed. Use of calculator is not allowed.
6. Figures to the right indicate full marks.
7. For each MCQ, it is mandatory to write the correct answer along with its alphabet. Only the first attempt will be considered for evaluation.

Physical Constants:

Mass of electron $m = 9.1 \times 10^{-31}$ kg

$\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/Nm^2$

$\pi = 3.142$

Charge on electron $e = 1.6 \times 10^{-19}$ C

$\mu_0 = 4\pi \times 10^{-7}$ Wb/Am

$h = 6.63 \times 10^{-34}$ Js

$c = 3 \times 10^8$ m/s

SECTION – A

Q.1 Select and write the correct answers: (10M)

(i) The first law of thermodynamics is concerned with conservation of

(a) momentum (b) energy (c) temperature (d) mass

(ii) The average value of alternating current over a full cycle is

(a) 0 (b) $I_0/2$ (c) $I_0/\sqrt{2}$ (d) $2I_0$

[I_0 = peak value of current]

(iii) The angle at which maximum torque is exerted by external uniform electric field on an electric dipole is

(a) 0° (b) 30° (c) 45° (d) 90°

(iv) Property of light which does not change when it travels from one medium to another is

(a) velocity (b) wavelength (c) frequency (d) amplitude

(v) Root mean square speed of gas molecules is proportional to

(a) T (b) $1/T$ (c) $\sqrt{T}$ (d) $1/\sqrt{T}$

[T = absolute temperature]

(vi) Unit $Wb\ m^{-2}$ is equal to

(a) henry (b) watt (c) dyne (d) tesla

(vii) For bob performing vertical circular motion, difference in tension between horizontal and top positions is

(a) mg (b) 2mg (c) 3mg (d) 6mg

(viii) Liquid rises 2.5 cm in a glass capillary. In another capillary of half radius, rise will be

(a) 1.25 cm (b) 2.5 cm (c) 5 cm (d) 10 cm

(ix) In Young's double slit experiment, if $I_{max} : I_{min} = 16 : 1$, ratio of amplitudes of two sources is

(a) 4 : 1 (b) 5 : 3 (c) 1 : 4 (d) 1 : 16

(x) Wave $y = 8 \sin (0.02x - 4t)$ cm has speed

(a) 10 cm/s (b) 20 cm/s (c) 100 cm/s (d) 200 cm/s

Q.2 Answer the following: (8M)

(i) Define potential gradient of potentiometer wire.

(ii) State formula for critical velocity in terms of Reynolds number.

(iii) Is it always necessary to use red light for photoelectric effect?

(iv) Write Boolean expression for Exclusive-OR (XOR) gate.

(v) Write differential equation for angular SHM.

(vi) What is mathematical formula for third postulate of Bohr's atomic model?

(vii) Two inductors 10 mH and 20 mH in series. Find resultant inductance.

(viii) Calculate MI of uniform disc of mass 10 kg, radius 60 cm about axis through centre perpendicular to plane.

SECTION – B

Attempt any Eight questions. (16M)

Q.3. Define moment of inertia of a rotating rigid body. State SI unit and dimensions.

Q.4. What are polar and non-polar dielectrics?

Q.5. What is a thermodynamic process? Name any two types.

Q.6. Derive expression for radius of nth Bohr orbit of electron in hydrogen atom.

Q.7. What are harmonics and overtones? (Two points.)

Q.8. Distinguish between potentiometer and voltmeter.

Q.9. What are mechanical equilibrium and thermal equilibrium?

Q.10. Electron orbits nucleus in circle of radius 5.3×10^{-11} m with speed 3×10^{6} m/s. Find angular momentum.

Q.11. Plane wave $\lambda = 6000$ Å on two slits, screen 2 m away. Distance between central bright and 10th bright is 2 cm. Find slit separation.

Q.12. Eight water droplets of radius 0.2 mm coalesce to form one drop. Find decrease in surface area.

Q.13. L = 0.1 H, C = 25×10^{-6} F, R = 15 Ω in series with 120 V, 50 Hz AC. Calculate resonant frequency.

Q.14. Difference $C_p - C_v = 9000$ J/kg·K and $C_p/C_v = 1.5$. Calculate Cp and Cv.

SECTION – C

Attempt any Eight questions. (24M)

Q.15. With neat diagram, explain reflection of light at a plane reflecting surface.

Q.16. What are magnetization, magnetic intensity and magnetic susceptibility?

Q.17. Prove beat frequency equals difference of frequencies of two notes.

Q.18. Define (a) inductive reactance (b) capacitive reactance (c) impedance.

Q.19. Derive expression for kinetic energy of a body rotating with uniform angular speed.

Q.20. Derive expression for emf induced in conductor of length l moving with velocity v in uniform magnetic field B.

Q.21. Derive expression for terminal velocity of a spherical body falling under gravity through viscous medium.

Q.22. Determine shortest wavelengths of Balmer and Paschen series given Lyman limit = 912 Å.

Q.23. Calculate magnetic field at 3 cm from long straight wire carrying 6 A.

Q.24. Parallel plate capacitor with air, area 6 cm², plate separation 3 mm. Calculate capacitance.

Q.25. Induced emf 91 mV when current in nearby coil rises at 1.3 A/s. Find mutual inductance in mH.

Q.26. Two cells 4 V (r = 1 Ω) and 2 V (r = 2 Ω) connected in parallel, both sending current in same direction through 5 Ω resistor. Find current through external resistance.

SECTION – D

Attempt any Three questions. (12M)

Q.27. Derive expression for pressure exerted by a gas using kinetic theory.

Q.28. What is a rectifier? With neat circuit diagram, explain working of half-wave rectifier.

Q.29. Draw neat labelled diagram of suspended coil type moving coil galvanometer.

Initial pressure and volume of gas: 2×10^5 N/m² and 6×10^{-3} m³. At constant pressure, 150 J work done in compression. Find final volume.

Q.30. Define seconds pendulum. Derive formula for its length.

A particle in SHM has vmax = 25 cm/s and amax = 100 cm/s². Find time period.

Q.31. Explain de Broglie wavelength. Obtain expression for de Broglie wavelength of material particle.

Photoelectric work function of metal is 4.2 eV. Find threshold wavelength.